As in [1], let (X, μ) be a probability space and consider a skew product $F\colon X^{\mathbb{N}} \times \mathbb{R} \circlearrowleft$ given by

$$F(\xi, y) = (\sigma(\xi), y + p(\xi_1))$$

for some $p \in L^1(X) \setminus \{0\}$ with average $\int p \, d\mu = 0$. Denote by $\mathbb{P} = \mu^{\otimes \mathbb{N}}$ the Bernoulli measure on $X^{\mathbb{N}}$.

Lemma 1 (false lemma)**.** *The map F is ergodic for the measure $\mathbb{P} \otimes \mathrm{Leb}$.*

The statement above is false. Using Hewitt-Savage we can show that any invariant A is such that $A_y = \{\xi : (\xi, y) \in A\}$ has measure 0 or 1 but this isn't enough. You may want to skip to the counterexample in red below.

Proof. Let $A \subseteq X^{\mathbb{N}} \times \mathbb{R}$ be a strictly invariant set. We claim that for fixed y, the set $A_y = \{\xi : (\xi, y) \in A\} \subseteq X^{\mathbb{N}}$ belong to the exchangeable σ-algebra of the projections $\{\pi_i\}_{i \in \mathbb{N}}$. Indeed, let $\alpha_{i,j}(\xi_1, \dots, \xi_i, \dots, \xi_j, \dots) = (\xi_1, \dots, \xi_j, \dots, \xi_i, \dots)$ be the action on $X^{\mathbb{N}}$ of a transposition exchanging $i \leftrightarrow j$, where $i \leq j$, and observe that since A is invariant, we can write

$$\begin{aligned}
A_y &= \{\xi : F^j(\xi, y) \in A\} \\
&= \{\xi : (\sigma^j(\xi), y + \cdots + p(\xi_i) + \cdots + p(\xi_j)) \in A\} \\
&= \{\xi : (\sigma^j(\alpha_{i,j}(\xi)), y + \cdots + p(\xi_j) + \cdots + p(\xi_i)) \in A\} \\
&= \{\xi : F^j(\alpha_{i,j}(\xi), y) \in A\} = \alpha_{i,j}(A_y).
\end{aligned}$$

By Hewitt-Savage, this means $\mathbb{P}(A_y) \in \{0, 1\}$.

Suppose $(\mathbb{P} \otimes \mathrm{Leb})(A) > 0$. By Fubini,

$$(\mathbb{P} \otimes \mathrm{Leb})(A^{\mathrm{c}}) = \int_{\mathbb{R}} \mathbb{P}(A_y^{\mathrm{c}}) \, d\,\mathrm{Leb}(y) = \mathrm{Leb}(\{y : \mathbb{P}(A_y) = 0\}).$$

Let's denote by $Z = \{y : \mathbb{P}(A_y) = 0\}$ and $P = \{y : \mathbb{P}(A_y) = 1\}$. From the assumption above, we know $\mathrm{Leb}(P) > 0$, and to conclude ergodicity, it suffices to show $\mathrm{Leb}(Z) = 0$.

Counterexample HERE

This is the point where it's impossible to finish the argument (and where the fact that $y \mapsto L(y)$ is increasing was crucial in [1]). Consider $X = \{-1, 1\}$, $\mu = \frac{1}{2}(\delta_{-1} + \delta_1)$ and $p(x) = x$ (as a real number, so the steps are either 1 or -1), and let

$$A = \bigcup_{k \in \mathbb{Z}} X \times \left[k, k + \frac{1}{2}\right].$$

It's simple to see A is invariant, because

$$\begin{aligned}
F^{-1}(A) &= \{(\xi, y) : \xi_1 = 1, (\sigma(\xi), y + 1) \in A\} \cup \{(\xi, y) : \xi_1 = -1, (\sigma(\xi), y - 1) \in A\} \\
&= \{(\xi, y) : \xi_1 = 1, \{y\} \in [0, 1/2]\} \cup \{(\xi, y) : \xi_1 = -1, \{y\} \in [0, 1/2]\} \\
&= A.
\end{aligned}$$

So I guess this is the reason that in [2] there are more hypothesis like ϕ smooth with two fixed points such that $\phi(p), \phi(q)$ are rationally independent... I think that in our case, just having something like

$$\forall \varepsilon > 0, \mu(\{x \in X : p(x) \in (-\varepsilon, \varepsilon)\}) > 0$$

should be enough? $\square$

References

[1] Araceli Bonifant and John Milnor. "Schwarzian Derivatives and Cylinder Maps". In: *American Mathematical Society* (2008), pp. 1–24.

[2] Sylvain Crovisier, Dawei Yang, and Jinhua Zhang. "Empirical Measures of Partially Hyperbolic Attractors". In: *Communications in Mathematical Physics* 375.1 (Apr. 1, 2020), pp. 725–764. ISSN: 1432-0916. DOI: 10.1007/s00220-019-03668-1. URL: https://doi.org/10.1007/s00220-019-03668-1 (visited on 03/25/2023).